

Additional file 1. Joint Effects for Interquartile Range Increases in 3-Day Moving Average of Multiple Ambient Air Pollutant Concentrations from Multipollutant Models without and with First Order Interactions.

Pollutant Combinations	Asthma or Wheeze		Pneumonia		Bronchitis		Otitis Media		URI	
	OR	95% CI	OR	95% CI	OR	95% CI	OR	95% CI	OR	95% CI
Oxidant Gases (O₃, NO₂, and SO₂)										
Without Interactions	1.032	(1.012, 1.052)	1.046	(1.017, 1.075)	1.049	(1.015, 1.083)	1.032	(1.020, 1.045)	1.053	(1.044, 1.063)
With Interactions	1.068 ^a	(1.040, 1.097)	1.044	(1.007, 1.081)	1.053	(1.015, 1.093)	1.038 ^a	(1.022, 1.054)	1.063 ^a	(1.050, 1.076)
Secondary Pollutants (O₃, SO₄²⁻, NO₃⁻, and NH₄⁺)										
Without Interactions	1.043	(1.019, 1.068)	1.068	(1.035, 1.102)	1.053	(1.019, 1.088)	1.025	(1.010, 1.039)	1.047	(1.036, 1.058)
With Interactions	1.042 ^a	(1.013, 1.072)	1.085 ^a	(1.047, 1.125)	1.090 ^a	(1.050, 1.132)	1.059 ^a	(1.042, 1.077)	1.063 ^a	(1.049, 1.076)
Traffic Pollutants (CO, NO₂, EC, and OC)										
Without Interactions	1.012	(0.999, 1.024)	1.009	(0.992, 1.026)	1.031	(1.010, 1.052)	1.018	(1.010, 1.026)	1.028	(1.022, 1.035)
With Interactions	1.022 ^a	(0.999, 1.046)	1.024	(0.995, 1.054)	1.038	(1.007, 1.071)	1.025 ^a	(1.012, 1.039)	1.043 ^a	(1.033, 1.054)
Coal Combustion Pollutants (SO₂ and SO₄²⁻)										
Without Interactions	1.026	(1.014, 1.038)	1.023	(1.007, 1.040)	1.013	(0.994, 1.032)	1.012	(1.004, 1.019)	1.020	(1.014, 1.025)
With Interactions	1.038 ^a	(1.024, 1.052)	1.019	(1.000, 1.037)	1.012	(0.993, 1.032)	1.012	(1.004, 1.020)	1.018	(1.012, 1.024)
Criteria Pollutants (O₃, CO, NO₂, SO₂, and PM_{2.5})										
Without Interactions	1.028	(1.008, 1.048)	1.041	(1.012, 1.070)	1.039	(1.005, 1.073)	1.031	(1.018, 1.043)	1.050	(1.040, 1.059)
With Interactions	1.055 ^a	(1.023, 1.089)	1.055	(1.013, 1.099)	1.051 ^a	(1.005, 1.099)	1.042 ^a	(1.022, 1.061)	1.063 ^a	(1.048, 1.078)

^a p-value of likelihood ratio test less than 0.05. The p-value was calculated from likelihood statistics based on chi-square distribution and the likelihood statistics were the change of -2log likelihood between models with and without interaction terms.